Density Waves in Fluids: Density, Momentum and Entropy Equations

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Outline

1. Dynamic Structure Factor $S(k, \omega)$
2. Shear Wave Attenuation
3. Critical Phenomena
4. Second Sound in Superfluids

1. Linearized Conservation Laws

The foundational hydrodynamic equations for mass, momentum, and energy, linearized for small-amplitude motions (primes denote fluctuations, subscript 0 denotes equilibrium).

- **Conservation of Mass (Continuity):**

$$\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \underline{v}' = 0$$

- **Conservation of Momentum:**

$$\frac{\partial \underline{v}'}{\partial t} + \frac{1}{\rho_0} \nabla p' - \frac{\eta}{\rho_0} \nabla^2 \underline{v}' - \frac{1}{\rho_0} \left(\zeta + \frac{1}{3}\eta \right) \nabla (\nabla \cdot \underline{v}') = 0$$

(Where p' is pressure, η is shear viscosity, and ζ is bulk viscosity.)

- **Conservation of Energy (Entropy):**

$$T_0 \rho_0 \frac{\partial s'}{\partial t} - k_{th} \nabla^2 T' = 0$$

2. Thermodynamic Coupling (Part 1)

To connect the four fluctuating variables (p', s', T', ρ'), we use exact thermodynamic relations:

$$p' = \left(\frac{\partial p}{\partial s}\right)_\rho s' + \left(\frac{\partial p}{\partial \rho}\right)_s \rho'$$

$$T' = \left(\frac{\partial T}{\partial s}\right)_\rho s' + \left(\frac{\partial T}{\partial \rho}\right)_s \rho'$$

The coefficients are defined by physical constants:

- Adiabatic speed of sound: $\left(\frac{\partial p}{\partial \rho}\right)_s = c^2$
- Specific heat at constant volume: $\left(\frac{\partial T}{\partial s}\right)_\rho = \frac{T_0}{c_v}$
- Maxwell relation: $\left(\frac{\partial T}{\partial \rho}\right)_s = \frac{1}{\rho_0^2}\left(\frac{\partial p}{\partial s}\right)_\rho$

2. Thermodynamic Coupling (Part 2)

Substituting these constants gives the **exact linearized hydrodynamic equations**:

- **Mass:** $\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \underline{v}' = 0$
- **Momentum:**

$$\frac{\partial \underline{v}'}{\partial t} + \frac{c^2}{\rho_0} \nabla \rho' + \rho_0 \left(\frac{\partial T}{\partial \rho} \right)_s \nabla s' - \frac{\eta}{\rho_0} \nabla^2 \underline{v}' - \frac{1}{\rho_0} \left(\zeta + \frac{1}{3} \eta \right) \nabla (\nabla \cdot \underline{v}') = 0$$

- **Entropy:**

$$\frac{\partial s'}{\partial t} - D_T \nabla^2 s' - \frac{c_p D_T}{T_0} \left(\frac{\partial T}{\partial \rho} \right)_s \nabla^2 \rho' = 0$$

Note: Thermal diffusivity is $D_T = \frac{k_{th}}{\rho_0 c_p}$ *and specific heat ratio is* $\gamma \equiv c_p / c_v \geq 1$. *Thermal expansion* β *couples these equations; if* $\beta = 0$, *the mechanical and entropy equations separate entirely.*

3. Spatial Fourier Transform & Velocity Splitting

Taking a spatial Fourier transform ($\nabla \to i\underline{k}$), the momentum equation becomes:

$$\frac{\partial \underline{v}'}{\partial t} + \frac{c^2}{\rho_0} i\underline{k}\rho' + \rho_0 \left(\frac{\partial T}{\partial \rho}\right)_s i\underline{k}s' + \frac{\eta}{\rho_0} k^2 \underline{v}' + \frac{1}{\rho_0}\left(\zeta + \frac{1}{3}\eta\right) \underline{k}(\underline{k} \cdot \underline{v}') = 0$$

Velocity is then separated into transverse $\underline{v}'_\perp$ and longitudinal v'_l components.

The Transverse Component (Shear Waves):
Uncoupled from density and entropy, simplifying to:

$$\frac{\partial \underline{v}'_\perp}{\partial t} + \nu k^2 \underline{v}'_\perp = 0$$

Where $\nu \equiv \eta/\rho_0$ is kinematic viscosity. These transverse shear waves are heavily attenuated with a complex wavenumber $\delta^{-1}(1+i)$, where $\delta = (2\nu/\omega)^{1/2}$ is the viscous penetration depth.

4. The Matrix & The Poles

Longitudinal velocity v_l' is coupled to density and entropy. A temporal Fourier transform solves the initial value problem, resulting in a 3×3 matrix.
Setting the determinant to zero gives the poles in the lower-half ω plane, relying on:

$$\frac{c_p \rho_0^2}{T_0 c^2} \left(\frac{\partial T}{\partial \rho} \right)_s^2 = \gamma - 1$$

Specific pole locations and widths:

- **Rayleigh Pole:** $\omega \approx -i\Gamma_R k^2$, where $\Gamma_R = D_T$.
- **Brillouin Poles:** $\omega \approx \pm ck - i\Gamma_B k^2$
  - Where $2\Gamma_B = D_l + (\gamma - 1)D_T$.
  - (Longitudinal viscous contribution: $D_l = \rho_0^{-1}(\zeta + \frac{4}{3}\eta)$).

5. The Final Spectral Denominators

The dynamic structure factor relies on the imaginary parts of the correlation functions $X''_{ij}(\underline{k}, \omega)$. Based on the roots, these take the form of Lorentzians with specific denominators:

- **Rayleigh Peak Denominator:**

$$\omega^2 + D_T^2 k^4$$

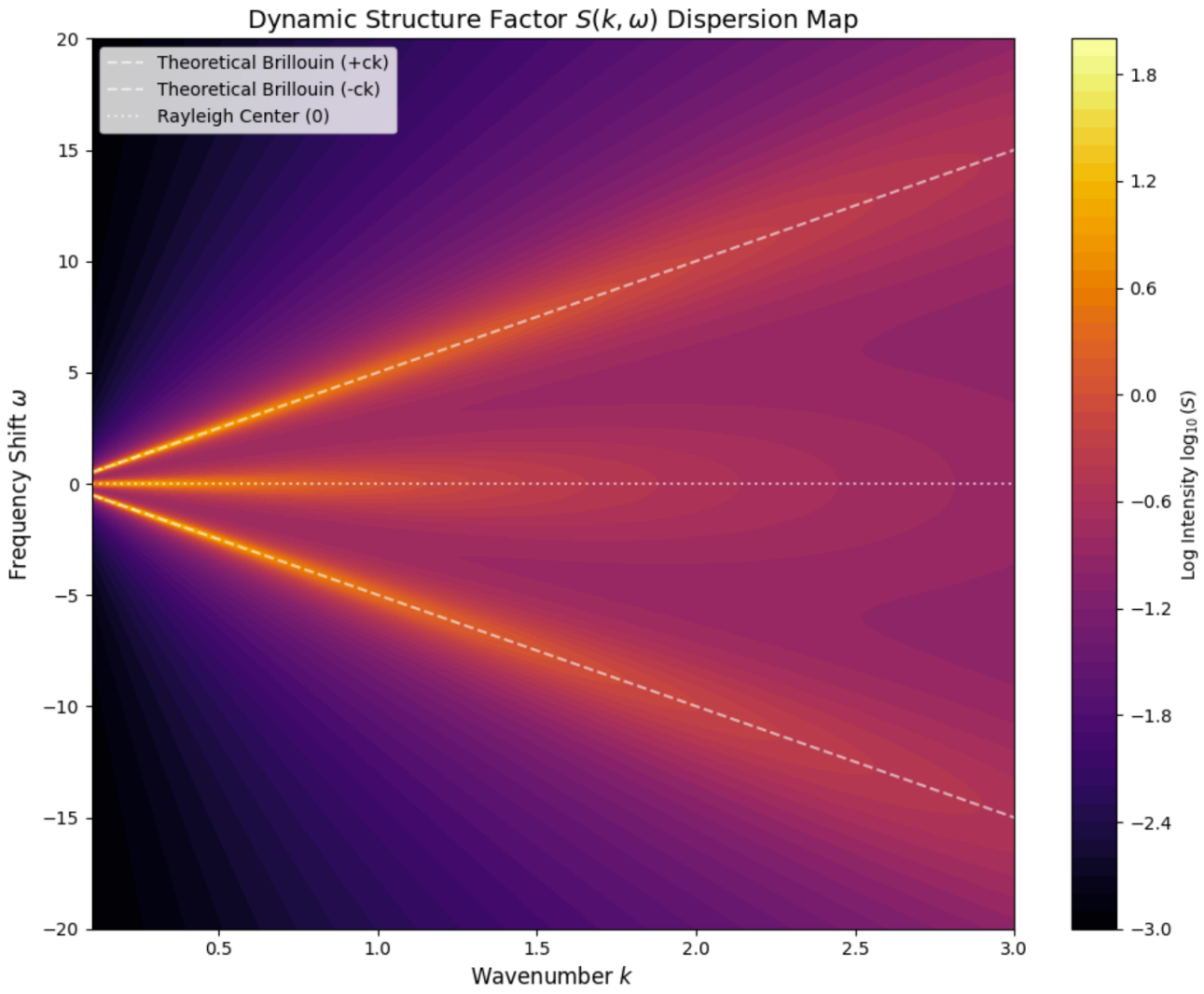
- **Brillouin Peaks Denominator:**

$$(\omega^2 - c^2 k^2)^2 + (2\omega k^2 \Gamma_B)^2$$

Resulting Intensity Profile (Fig 10.1):
This shapes the profile into a central diffusive Rayleigh peak (width $D_T k^2$) and a pair of Brillouin peaks (width $\Gamma_B k^2$) shifted by $\pm ck$.

Dynamic Structure Factor $S(k, \omega)$ Dispersion Map

For c $= 5.0$, $D_T = 0.80$, $D_l = 0.50$ and $\gamma = 1.4$



Conclusions

Based on the provided dispersion map of the Dynamic Structure Factor $S(k, \omega)$, we can draw several powerful physical inferences about how fluctuations behave in this fluid across different length scales:

- **Linear Dispersion of Sound:** The brightest diagonal bands perfectly track the theoretical $\omega = \pm ck$ dashed lines. This confirms that density fluctuations propagate as true acoustic waves. Because the relationship is strictly linear, we can infer that the speed of sound c (the slope of the line) remains constant across all the observed wavenumbers, without anomalous dispersion in this regime.
- **Diffusive Nature of Entropy:** The central, horizontal band remains permanently anchored at a frequency shift of $\omega = 0$. This visually confirms that thermal/entropy fluctuations do not travel as waves (they have zero propagation speed) but simply sit in place and diffuse heat into the surrounding fluid.
- **Wavenumber-Dependent Damping (k^2 Broadening):** If you look at the left side of the graph (low k), the peaks are very sharp and distinct (bright yellow). As you move to the right (higher k, representing shorter wavelengths), the bands become much wider, fainter, and start to blur together into the red/purple regions. This is a direct visual proof of the theory we discussed earlier: the damping widths ($\Gamma_R k^2$ and $\Gamma_B k^2$) scale quadratically. It tells us physically that short-wavelength, microscopic ripples are killed off by the fluid's viscosity and thermal conductivity much faster than large, macroscopic waves.

1. Vector Splitting in Fourier Space

While longitudinal waves (sound and heat) dominate the light-scattering spectrum, hydrodynamic equations also predict **transverse motions** (shear stresses where fluid layers slide past one another).

After applying a spatial Fourier transform, we separate the velocity vector v' into two orthogonal parts relative to the wavevector $\underline{k}$:

- **Longitudinal component (v_l'):** Motion parallel to wave propagation (compression/expansion).
- **Transverse component ($\underline{v}_\perp'$):** Motion perpendicular to wave propagation (shearing).

Mathematical Separation:

$$v'(\underline{k}, t) = \underline{v}_\perp'(\underline{k}, t) + \hat{k} v_l'(\underline{k}, t)$$

2. The Uncoupled Transverse Equation

Projecting the momentum equation onto the transverse direction simplifies everything. Pressure ($\nabla p'$) and entropy ($\nabla s'$) gradients are purely longitudinal.

The transverse velocity is completely uncoupled from density and temperature fluctuations:

$$\frac{\partial \underline{v}'_\perp}{\partial t} + \nu k^2 \underline{v}'_\perp = 0$$

(Where $\nu \equiv \eta/\rho_0$ is the kinematic viscosity.)

Analogy to Spin Diffusion:
This has the exact same mathematical form as the diffusion of magnetization.

- Transverse momentum **diffuses** through the fluid exactly like heat or spin.
- It does **not** oscillate or propagate as a free wave.

3. Spatial Attenuation & Penetration Depth

If we force these waves to propagate (e.g., mechanically oscillating a plate), viscosity resists the motion, heavily attenuating the waves.

Complex Wavenumber:
The waves take on a complex wavenumber, where the imaginary part causes rapid exponential decay:

$$k_{\text{complex}} = \delta^{-1}(1 + i)$$

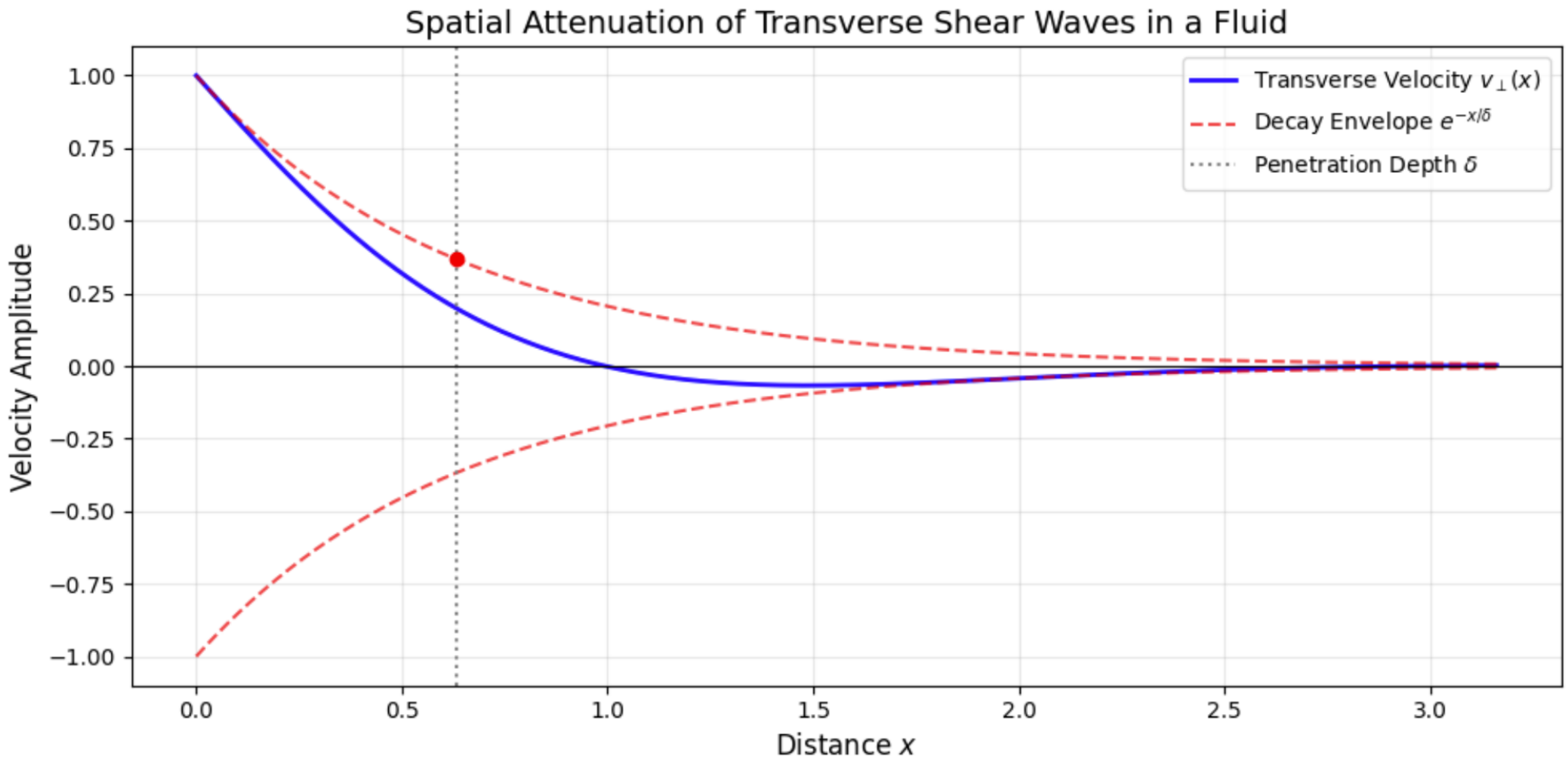
Viscous Penetration Depth (δ):

$$\delta = \left(\frac{2\nu}{\omega}\right)^{1/2}$$

- **High frequencies (ω):** Smaller $\delta \implies$ fast vibrations die out almost immediately at the surface.
- **High kinematic viscosity (ν):** Larger $\delta \implies$ "stickier" fluids transfer shear force deeper into the bulk before dissipating.

Spatial Attenuation of Transverse Shear Waves in a Fluid

$\nu = 1.0$ and $\omega = 5.0$



Conclusions

Based on the plot showing the "Spatial Attenuation of Transverse Shear Waves in a Fluid," we can draw a few striking physical inferences about how fluids handle shear forces:

- **Extreme Spatial Damping:** Unlike sound waves (longitudinal waves) that can travel across an entire room or ocean, this transverse shear wave dies out almost immediately. The solid blue line shows that the fluid's velocity amplitude crashes toward zero over a very short distance. This visually proves why we don't observe propagating shear waves in normal fluids—they are heavily and rapidly absorbed by internal friction (viscosity).
- **The Significance of the Penetration Depth (δ):** The vertical dotted line marks $x = \delta$. Notice exactly where the red dot sits on the decay envelope at this line: it is at an amplitude of roughly 0.37 (which is $1/e$). This confirms that δ isn't just an arbitrary marker; it is the specific characteristic length where the wave has lost about 63% of its initial energy. Beyond a distance of 3δ to 4δ, the fluid is completely undisturbed.
- **Locked Oscillation and Decay:** Look at the oscillation itself (the blue line). It doesn't even complete a single full cycle before it is completely squashed by the red decay envelope. This happens because, as derived earlier, the complex wavenumber is proportional to $1 + i$. This mathematically locks the spatial oscillation and the exponential decay to the exact same length scale (δ). The fluid kills the wave just as fast as it tries to oscillate.

Critical Phenomena & Opalescence

When a fluid approaches its **liquid-gas critical point** (the temperature T_c where the distinction between liquid and gas disappears), the specific heat at constant pressure, c_p, **diverges** (approaches infinity).

1. Critical Opalescence (The Rayleigh Peak Dominates)
The ratio of the area under the central Rayleigh peak to the Brillouin doublet is roughly $\gamma - 1$, where $\gamma = c_p/c_v$.

- **The Math:**
  As $T \to T_c$, $c_p \to \infty$. Therefore, $\gamma \to \infty$.
- **The Physics:**
  The Rayleigh peak (entropy fluctuations) absorbs almost all the scattered light intensity. The massive scattering intensity causes a normally clear fluid to suddenly look milky and opaque (**critical opalescence**).

2. Critical Slowing Down (The Rayleigh Peak Shrinks)
The width of the Rayleigh peak is determined by the thermal diffusivity:

$$D_T = \frac{k_{th}}{\rho_0 c_p}$$

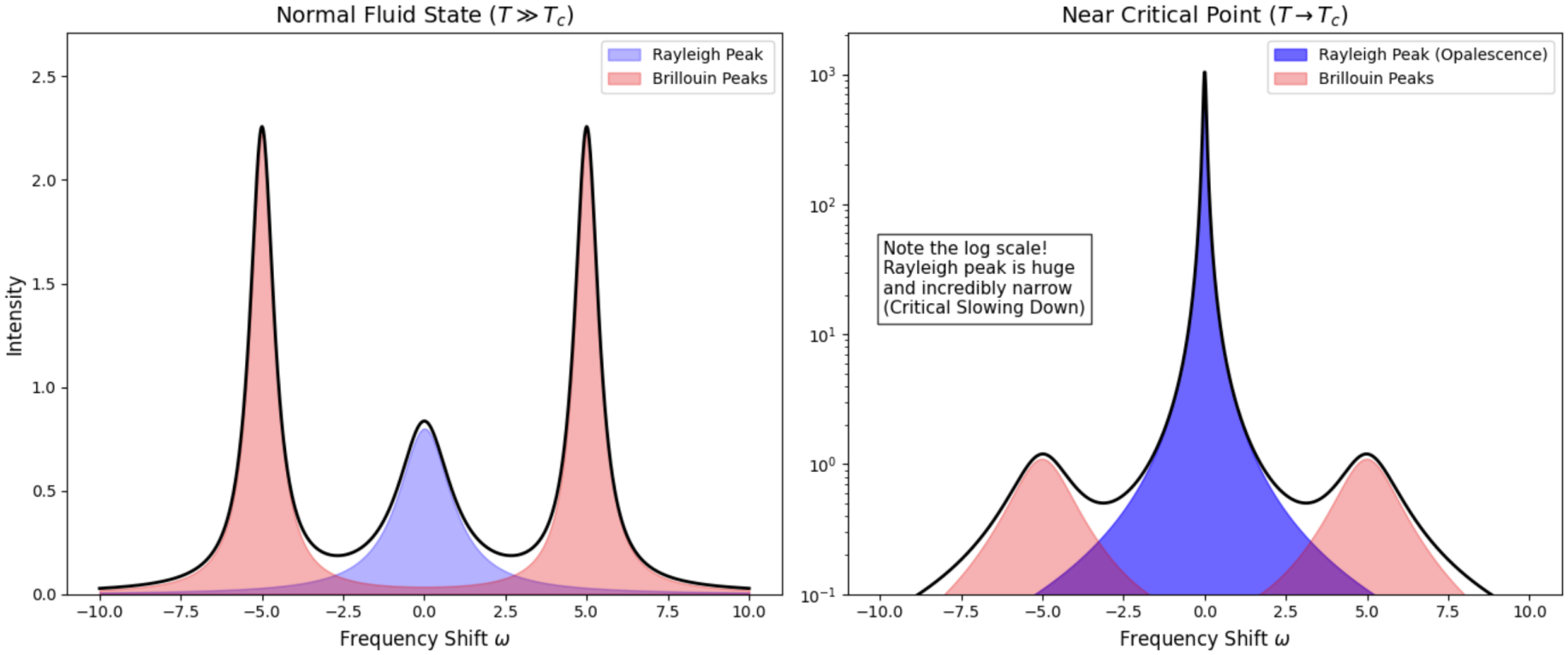
- **The Math:**
  Because c_p is in the denominator, as $c_p \to \infty$, $D_T \to 0$. Since the width of the Rayleigh peak is $D_T k^2$, the peak becomes **infinitely narrow**.

- **The Physics:**
  A narrow peak in the frequency domain (ω) corresponds to a very slow decay in the time domain (t). Entropy and temperature fluctuations take a very long time to dissipate (**critical slowing down**).

Note: Modern theories use power-laws based on $T - T_c$ to model exactly how fast c_p diverges, and light scattering has been central to verifying these critical exponents.

Critical Phenomena

k = 1.0, c = 5.0, $D_l = 0.50$, $\gamma_{normal} = 1.40$, $D_{T(normal)} = 1.0$ and C_p Divergence Factor = 20



Conclusions

Based on the side-by-side comparison of the fluid in a normal state versus near its critical point, we can draw some profound physical inferences about phase transitions:

- **Massive Divergence in Intensity (Critical Opalescence):** Normally, Brillouin and Rayleigh peaks are comparable. Approaching the critical point, the Rayleigh peak overwhelmingly dominates—a massive disparity masked by the logarithmic scale. This occurs because specific heat (c_p) approaches infinity; the resulting entropy fluctuations scatter intense light, turning a transparent fluid milky and opaque.
- **The Collapse of Thermal Diffusion (Critical Slowing Down):** Notice how wide and gentle the blue Rayleigh peak is in the normal state, but how incredibly sharp and needle-like it becomes near the critical point. Since the width of this peak represents thermal diffusivity (D_T), a near-zero width means D_T has essentially collapsed. Physically, this tells us that the fluid has lost its ability to efficiently diffuse heat; microscopic temperature and entropy fluctuations get "stuck" and take an extraordinarily long time to dissipate back into equilibrium.
- **Sound Waves are Dwarfed, Not Destroyed:** If you look closely at the right graph, the red Brillouin peaks haven't disappeared—they are still sitting there at $\pm\omega = 5.0$. However, because the entropy fluctuations (the blue peak) have become completely unhinged and grown by orders of magnitude, the acoustic density waves simply become insignificant contributors to the total light scattering.

The Superfluid Transition

In a **normal fluid**, heat spreads out slowly via diffusion.

- This is why the Rayleigh peak sits stationary at $\omega = 0$ and just broadens.

Liquid Helium-4 and the Lambda Point:
When Liquid Helium-4 is cooled below a critical temperature called the **Lambda point** ($T_\lambda \approx 2.17$ Kelvin), it undergoes a quantum phase transition into a **superfluid**.

- A superfluid is a bizarre quantum state characterized by having **zero viscosity**.

In this quantum superfluid state, heat no longer diffuses. Instead, **heat propagates as a wave**.

This thermal wave is famously called **"Second Sound"**.

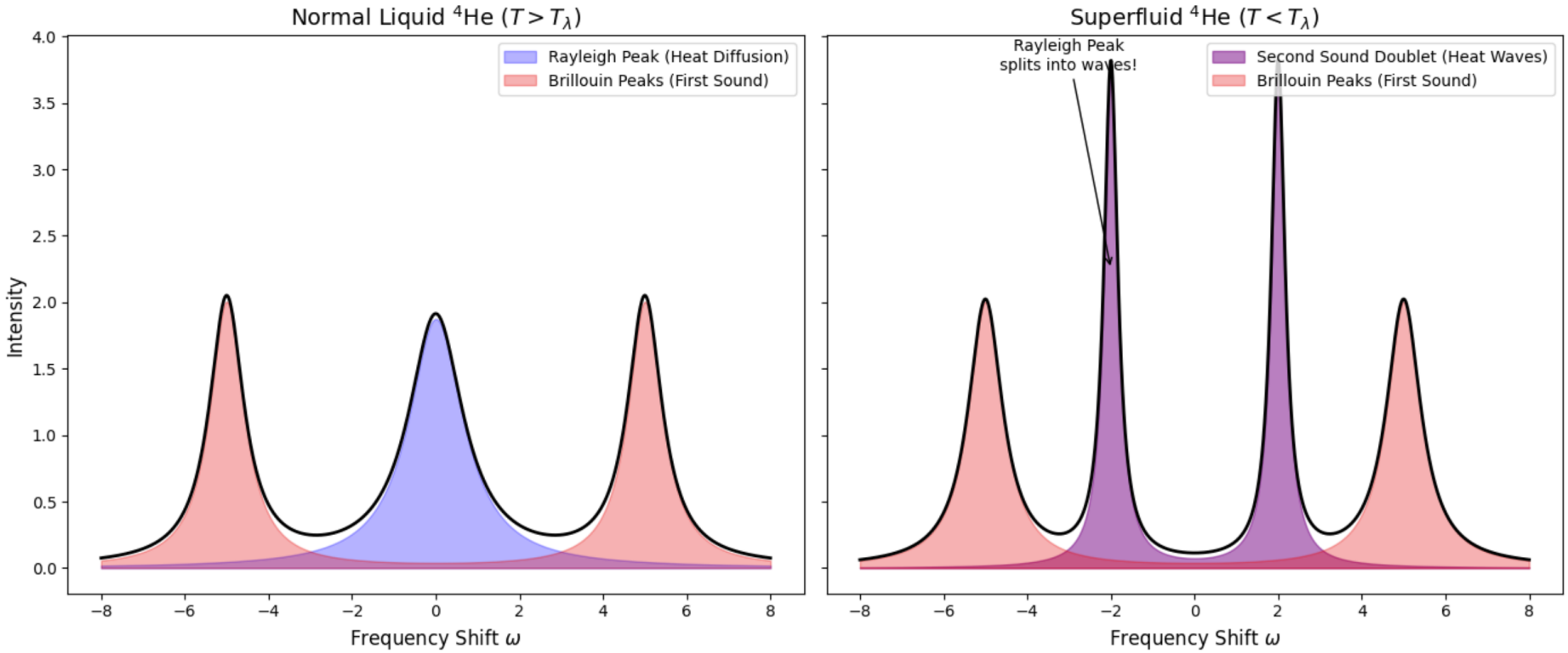
Impact on the Light Scattering Spectrum:
Because heat is now traveling like a sound wave, the spectrum undergoes a dramatic transformation:

- The central Rayleigh peak (entropy fluctuations) **completely disappears**.
- It **splits into a new pair of peaks** (a doublet) located at $\pm c_2 k$, where c_2 is the speed of second sound.

Note: The paper notes this conceptually but leaves the explicit math to "Ref. 8, Chap. 10". Even without the exact matrix, we can clearly model what this spectrum looks like conceptually!

Second Sound in Superfluids

$k = 1.0,\ c_1 = 5.0,\ c_2 = 2.0,\ w_R = 0.8,\ w_B = 0.5,\ w_S = 0.2,\ A_B = 1.0,\ A_R = 1.5,$
$A_S = \frac{A_R}{2}$



Conclusions

Based on the side-by-side comparison of Normal Liquid ^{4}He and Superfluid ^{4}He, we can draw some remarkable physical inferences about quantum phase transitions:

- **A Quantum Shift in Heat Transport:** The most profound feature of this graph is the complete disappearance of the central blue peak at $\omega = 0$ in the right panel. This visually proves that in a superfluid, normal heat diffusion ceases to exist.Instead, the Rayleigh peak splits into a purple doublet, confirming that temperature and entropy fluctuations now propagate as true physical waves—a phenomenon known as "second sound".
- **Two Distinct Wave Speeds:** The graph shows two sets of propagating waves in the superfluid state. Because the purple "second sound" peaks ($\pm\omega \approx 2$) are closer to the center than the red "first sound" Brillouin peaks ($\pm\omega \approx 5$), we can infer that these thermal waves travel slower than standard acoustic pressure waves.
- **Frictionless Propagation:** Notice how sharp and narrow the purple second sound peaks are compared to the broad, mound-like Rayleigh diffusion peak on the left. This extreme sharpness indicates that the thermal waves experience very little damping or attenuation. This lack of resistance is a direct visual signature of the superfluid's frictionless, zero-viscosity quantum state.
- **Coexistence of Fluids (Two-Fluid Model):** The fact that the red Brillouin peaks (first sound/density waves) remain largely unchanged in the right graph shows that standard acoustic waves still propagate through the medium alongside the new quantum heat waves.

Thank You!